A Foot-Controlled Sixth Finger Augmentation Interface

1. Design Idea and Motivation

This report proposes a foot-controlled wearable sixth finger as a motor augmentation interface to enhance hand manipulation capabilities. The idea is to add an additional controllable finger to the hand, allowing the user to stabilise and manipulate objects more effectively during every day and task-oriented activities.

I chose this idea because many daily tasks implicitly require more than two points of contact, such as holding a bottle while opening a cap, stabilising an object while using a tool, or carrying items while interacting with doors or touch interfaces. In these situations, users often rely on their second hand or external support. A sixth finger offers an opportunity to extend the hand's functional workspace without replacing or restricting natural hand movements.

A major challenge in augmentation systems is the design of an intuitive control interface. Using the hand itself as the control source may interfere with the task being augmented. For this reason, this design explores foot-based control, exploiting the fact that the feet are often idle during upper-limb manipulation tasks. Foot control also provides a spatially separated input channel, reducing interference with natural hand coordination.

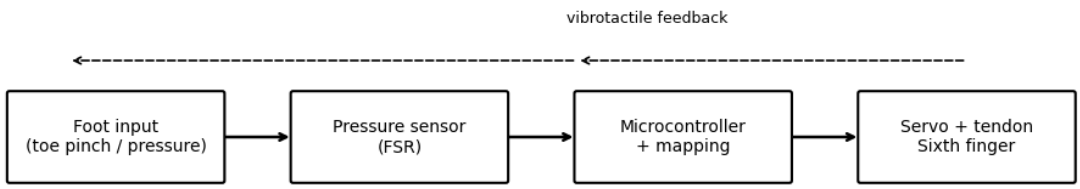

Figure 1. Concept illustration of the foot-controlled sixth finger worn on the hand

2. Function Augmented and Augmentation Principle

2.1 Augmented function

The proposed system performs motor augmentation, specifically degree-of-freedom (DoF) augmentation. By adding an extra finger, the hand gains an additional controllable contact point that can improve grasp stability and task efficiency.

The sixth finger does not replace any existing finger or restore lost function. Instead, it extends the natural capabilities of the hand by enabling more stable grasps on large or awkward objects and reduced need for grip re-adjustments during manipulation.[1][2]

2.2 Augmentation Interface Concept

The augmentation interface forms a closed-loop interaction between the user and the device:

- **Input**: pressure or pinching action using the toes
- **Mapping**: conversion of foot pressure into finger motion
- **Output**: flexion of the sixth finger to assist grasping

Foot pressure is continuously mapped to the bending angle of the sixth finger, allowing proportional rather than binary control. This mapping supports gradual learning and integration into the user's natural motor behaviour.

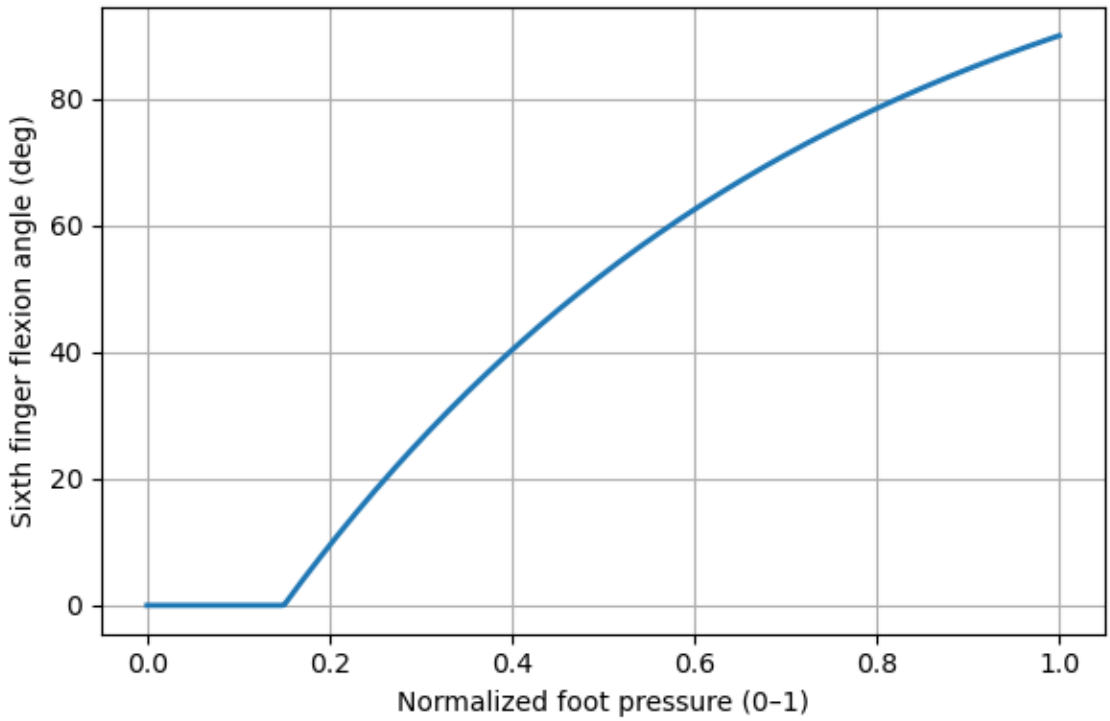

Figure 2. Control mapping diagram from foot pressure input to sixth finger flexion

3. Interface Design and System Description

3.1 Six Finger Mechanism

The sixth finger is mounted on the side of the hand near the little finger or along the palm edge. It consists of a lightweight articulated structure with one degree of freedom, designed to flex towards the palm. Actuation is achieved using a small servo motor connected via a tendon-driven mechanism.

A soft or compliant fingertip is included to increase friction and ensure safe interaction with objects. The finger is intended to provide stabilising support rather than precise independent manipulation.

3.2 Foot-based Control Interface

The control interface is implemented using force-sensitive resistors (FSRs) mounted in a foot-worn attachment. Two design options are considered:

- A toe-pinch interface using sensors placed between the big toe and second toe
- A pressure pad under the forefoot activated by downward force

The sensor signal is processed by a microcontroller and mapped to the servo motor position. Dead zones and saturation limits are implemented to prevent accidental activation and excessive gripping force.

Figure 3. Foot-mounted pressure sensing interface

3.3 Feedback and Usability

To improve usability, simple feedback can be provided through a vibration motor placed on the hand or foot. This feedback indicates when sufficient grip force has been achieved, reducing reliance on visual attention and supporting more natural interaction with the augmented finger.

4. Development Plan

The system would be developed through iterative prototyping:

- Validation of foot pressure sensing and signal stability
- Characterisation of finger actuation and mechanical reliability

- Integration into a wearable hand-mounted structure
- Tuning of control mapping and response speed
- Informal user testing during simple manipulation tasks

Performance could be evaluated by comparing task completion time, grip failures, and subjective workload with and without the sixth finger.

5. Limitations and Future Work

Foot-based control may be less suitable during walking or dynamic postures, and care must be taken to avoid unintended activation. Additionally, FSR sensors suffer from drift and limited repeatability, which may affect long-term robustness.

Future work could explore alternative sensing modalities for the foot, multi-joint or soft robotic finger designs, and adaptive control strategies that learn user-specific input patterns over time.

6. Conclusion

This design presents a foot-controlled sixth finger augmentation interface that extends hand functionality by adding an extra controllable degree of freedom. By separating control input from the augmented limb, the interface minimises interference with natural hand movements while enabling more stable and efficient object manipulation. The concept aligns with principles of motor augmentation and human-centred design and represents a feasible direction for wearable human augmentation systems.

Reference

[1] J. Eden, C. Mehring, and E. Burdet, “Principles of movement augmentation,” in *Proceedings of the International IEEE Conference on Neural Engineering (NER)*, San Diego, CA, USA, 2021

[2] C. Mehring, J. Eden, O. d. Xivry, E. Burdet, and K. Diedrichsen, “Augmented manipulation ability in humans with six-fingered hands,” *Nature Communications*, vol. 10, no. 1, pp. 1–10, 2019.